

Overview of Artificial Intelligence (AI)

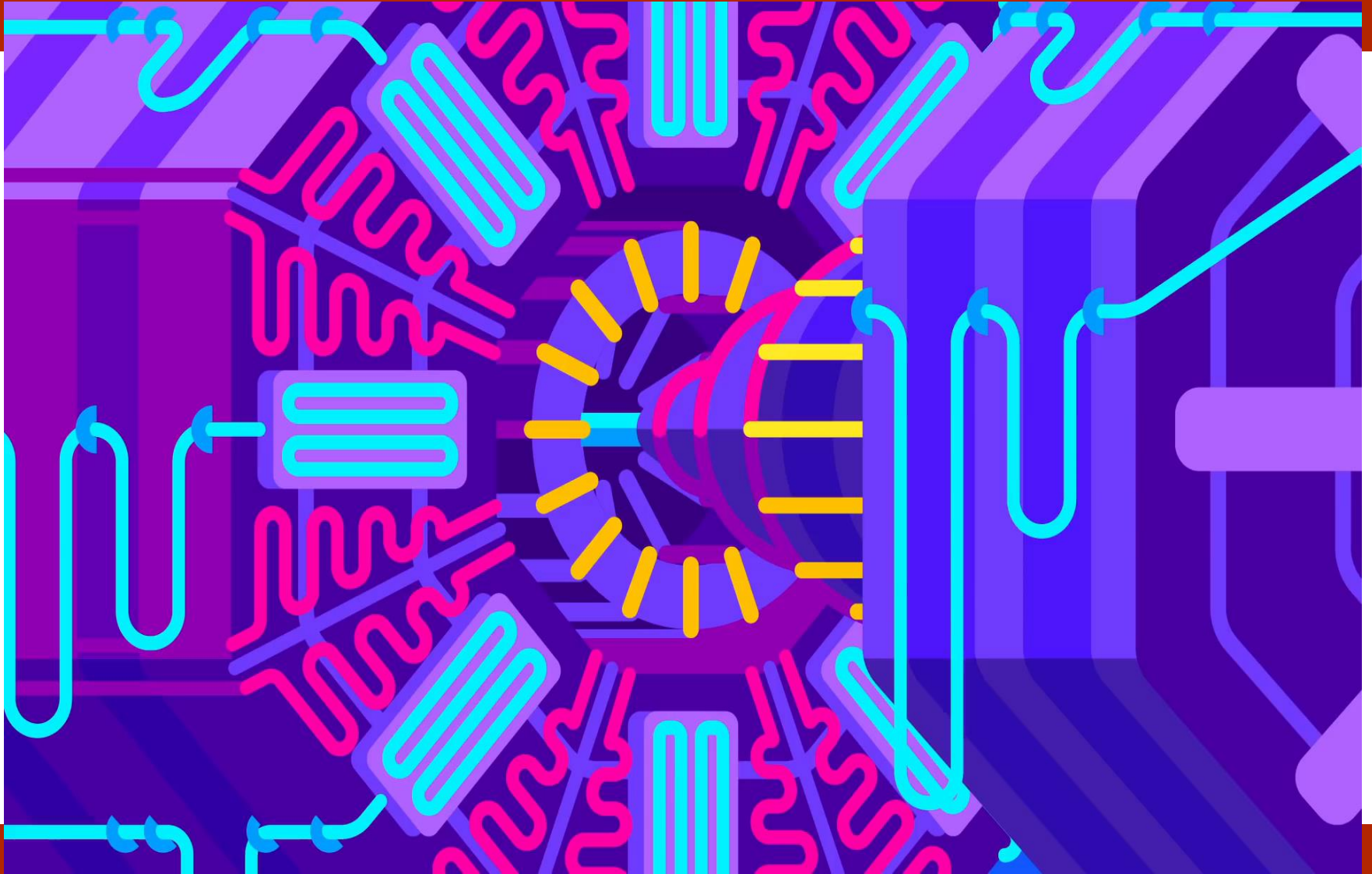
By: Faisal Hussain

Agenda

❑ **Foundations of AI & Machine Learning**

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Deep Learning (DL)
- Types of Machine Learning
- Types of Data
- Basic Terminologies of Machine Learning

What is Intelligence?

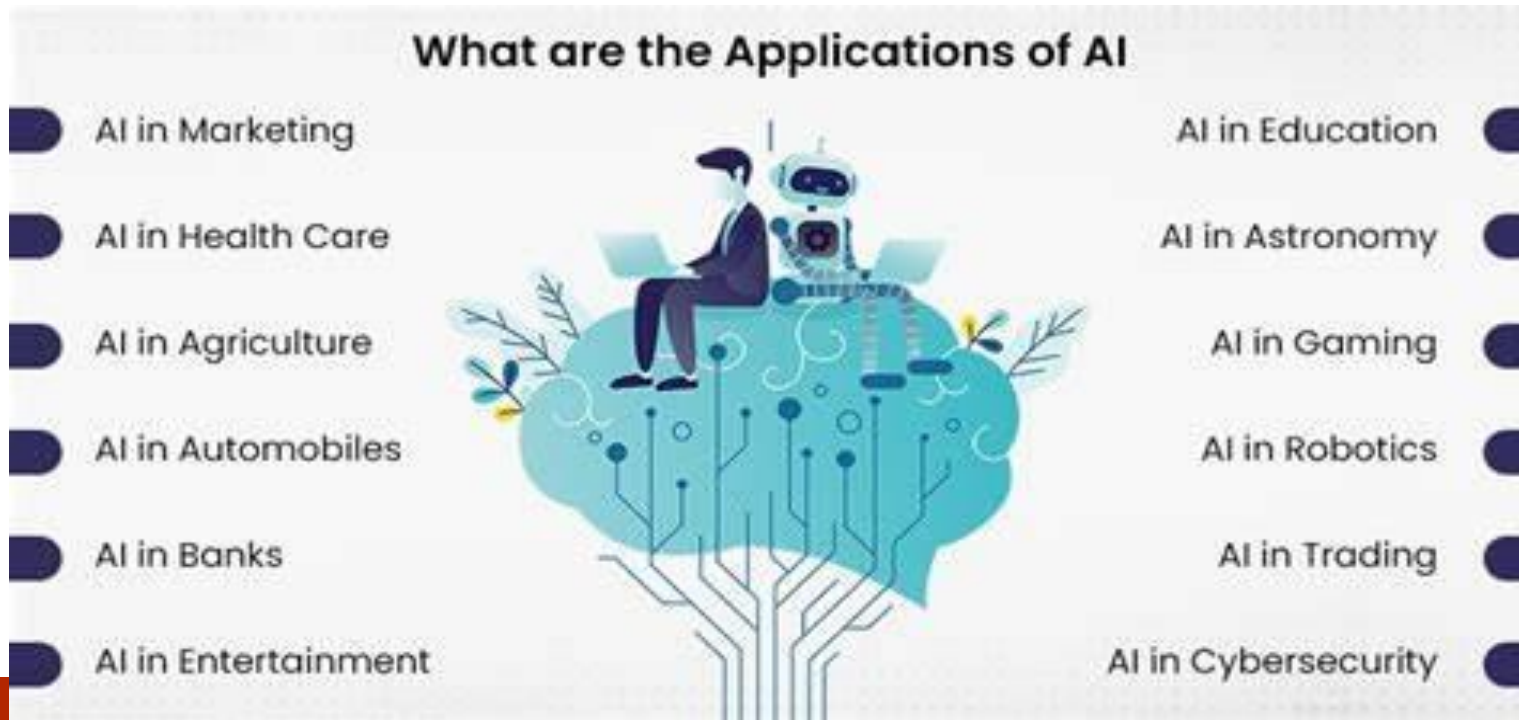


What is Artificial Intelligence?

- **What is intelligence?**
 - Intelligence is a mechanism to solve problems.
 - Includes the ability to gather knowledge, to learn, be creative, form strategies, or engage in critical thinking.

What is Artificial Intelligence?

- Artificial intelligence leverages computers and machines to mimic the problem-solving and decision-making capabilities of the human mind.



Types of Artificial Intelligence

Artificial Narrow Intelligence (ANI)



Stage-1

Machine Learning

- Specialises in one area and solves one problem



Siri



Alexa



Cortana

Artificial General Intelligence (AGI)



Stage-2

Machine Intelligence

- Refers to a computer that is as smart as a human across the board

Artificial Super Intelligence (ASI)



Stage-3

Machine Consciousness

- An intellect that is much smarter than the best human brains in practically every field

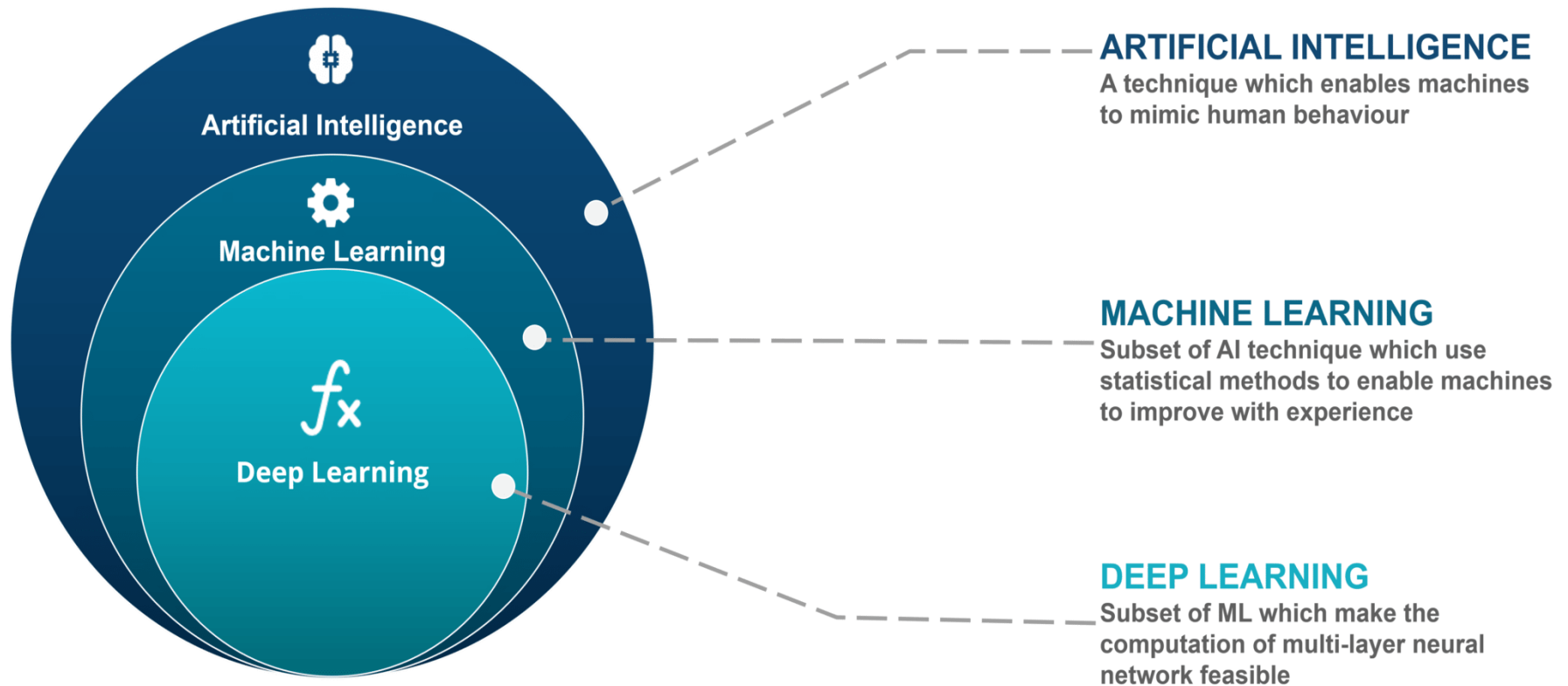
What is Machine Learning (ML)?

Machine learning is a branch of artificial intelligence (AI) that allows the system to automatically learn and improve from experience **without being explicitly programmed.**

What is Deep Learning?

- Deep learning is actually comprised of **neural networks**.
- “Deep” in deep learning refers to a neural network comprised of more than three layers—which would be inclusive of the inputs and the output—can be considered a deep learning algorithm.

AI vs ML vs DL - What's the difference?



Types of Machine Learning

- Supervised Learning
- Unsupervised Learning
- Reinforcement learning

Supervised Learning:

- Train the machine with labeled data.
- Then we test our model with some unknown new set of data and predict the level for them.

Types of Machine Learning

Unsupervised learning

- It mainly deals with the unlabeled data.
- You need to allow the model to work on its own to discover information.

Reinforcement Learning:

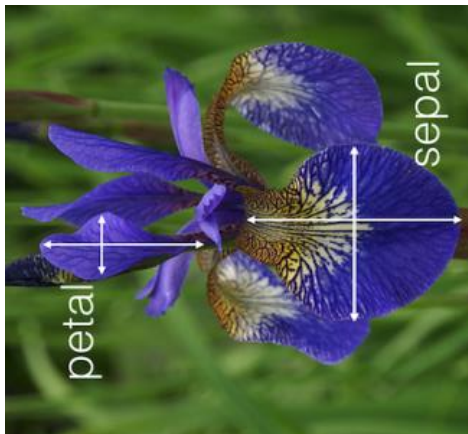
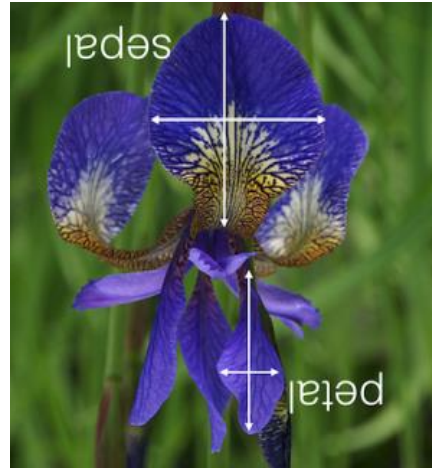
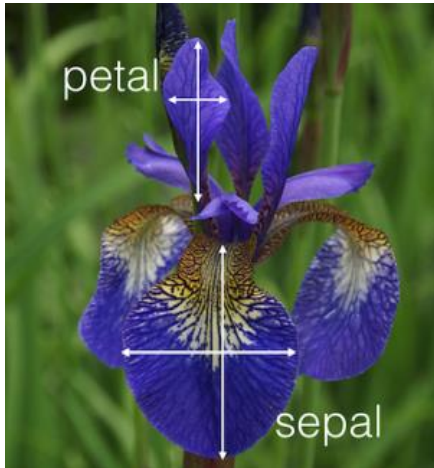
- Reinforcement learning is about taking suitable action to maximize reward in a particular situation.

Machine Learning Basic Terminologies

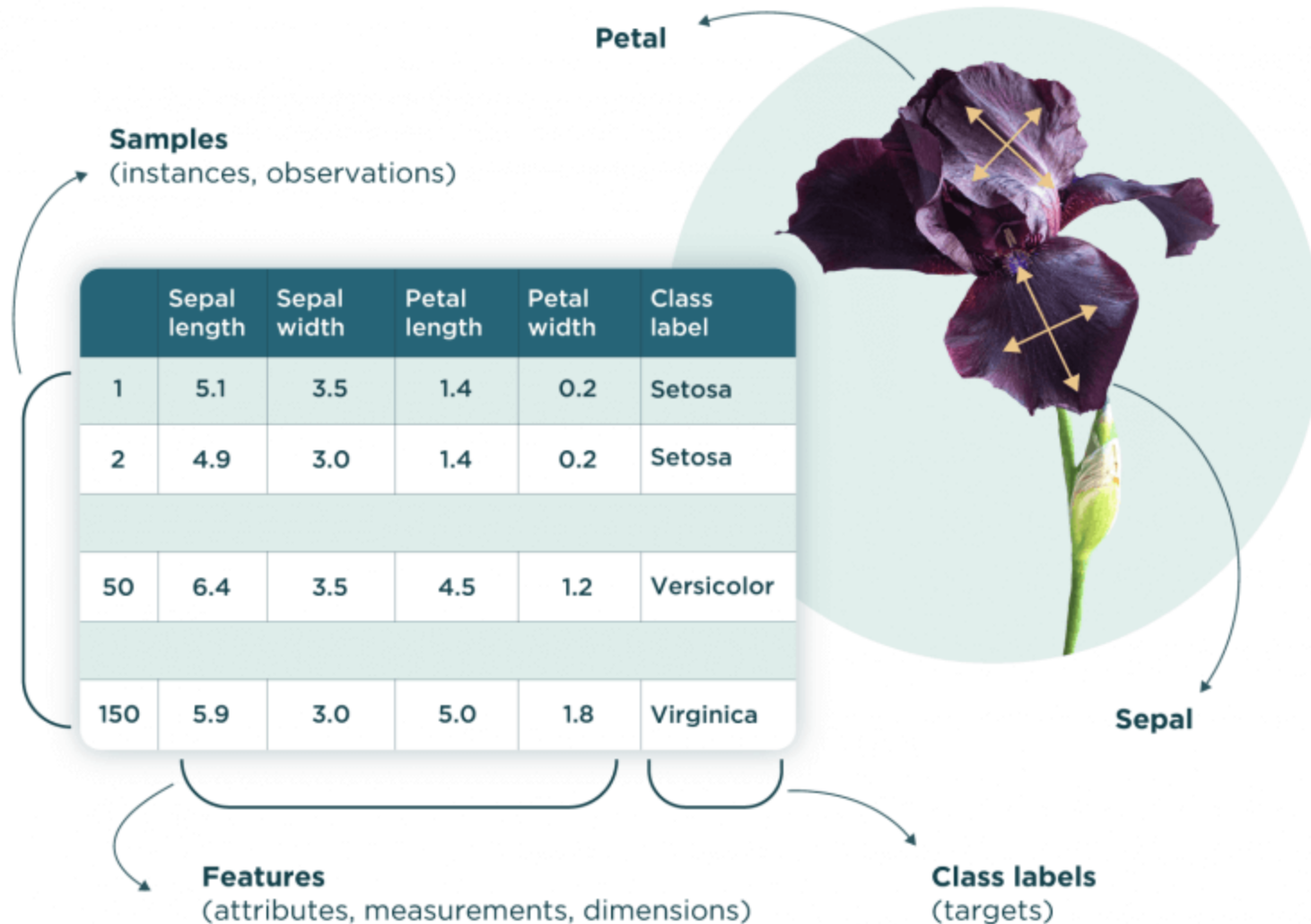
Data & Types of Data

- **Dataset** → A collection of observations, instances, samples
- **Sample/Example** → One row of a dataset.
- A sample contains **features** and possibly a **label**.
- Labelled/Unlabeled Dataset
- Structured vs Unstructured Data
- Categorical Data, Numerical Data
- Discrete Values, Continuous Values

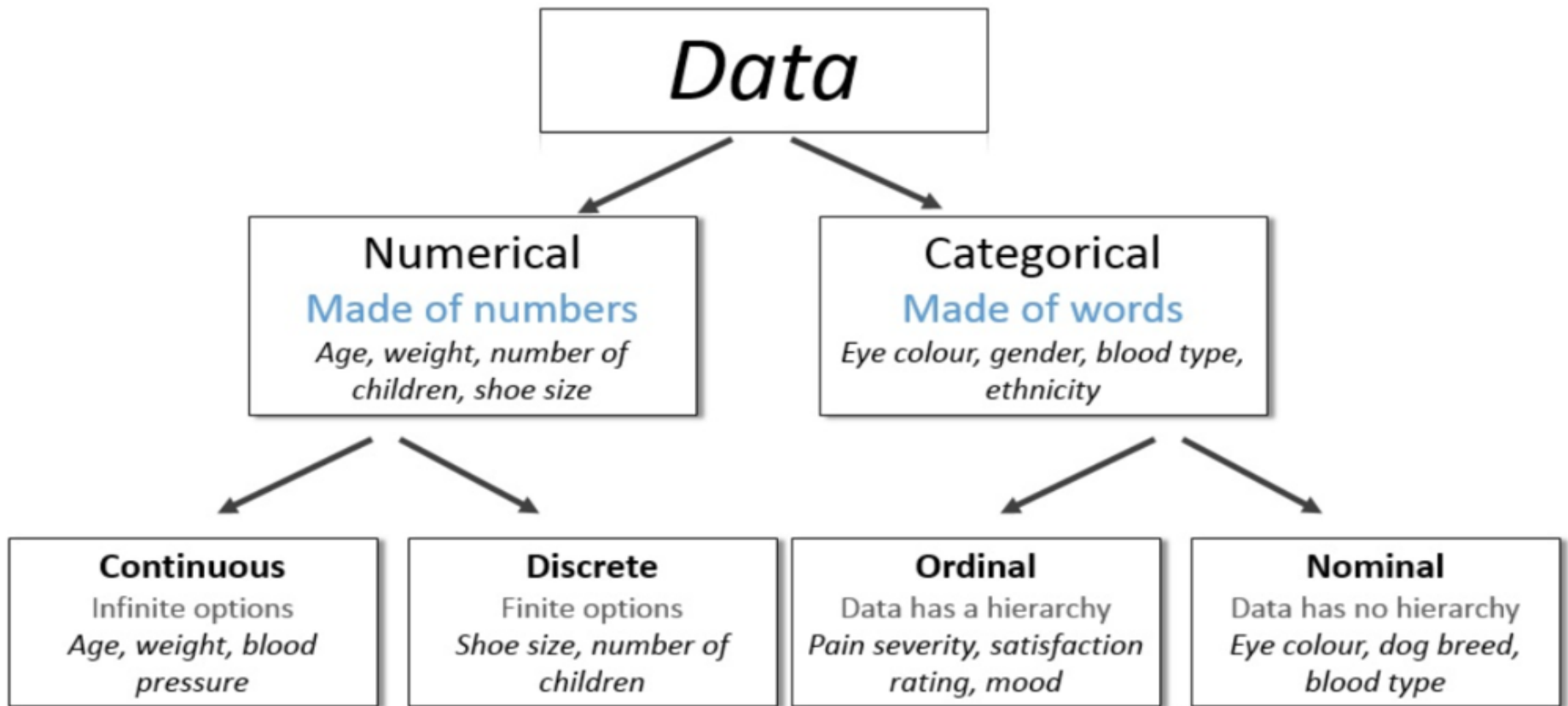
Unstructured Data (image, audio, video, text)



Structured Data (.csv, .txt)



Machine Learning Basic Terminologies



Machine Learning Basic Terminologies

Features

- An input variable used in making predictions.

Features Set

- The group of features your machine learning model trains on.
- For example, [property size](#), and [property condition](#) might comprise a simple feature set for a model that [predicts housing prices](#).

Features Extraction

- Retrieving intermediate features for better model training.

Features Selection

- Selecting the useful features for model training using various statistical or machine learning algorithms.

Machine Learning Basic Terminologies



Position	Experience	Skill	Country	City	Salary (\$)
Developer	0	1	USA	New York	103100
Developer	1	1	USA	New York	104900
Developer	2	1	USA	New York	106800
Developer	3	1	USA	New York	108700
Developer	4	1	USA	New York	110400
Developer	5	1	USA	New York	112300
Developer	6	1	USA	New York	114200
Developer	7	1	USA	New York	116100
Developer	8	1	USA	New York	117800
Developer	9	1	USA	New York	119700
Developer	10	1	USA	New York	121600

Machine Learning Basic Terminologies

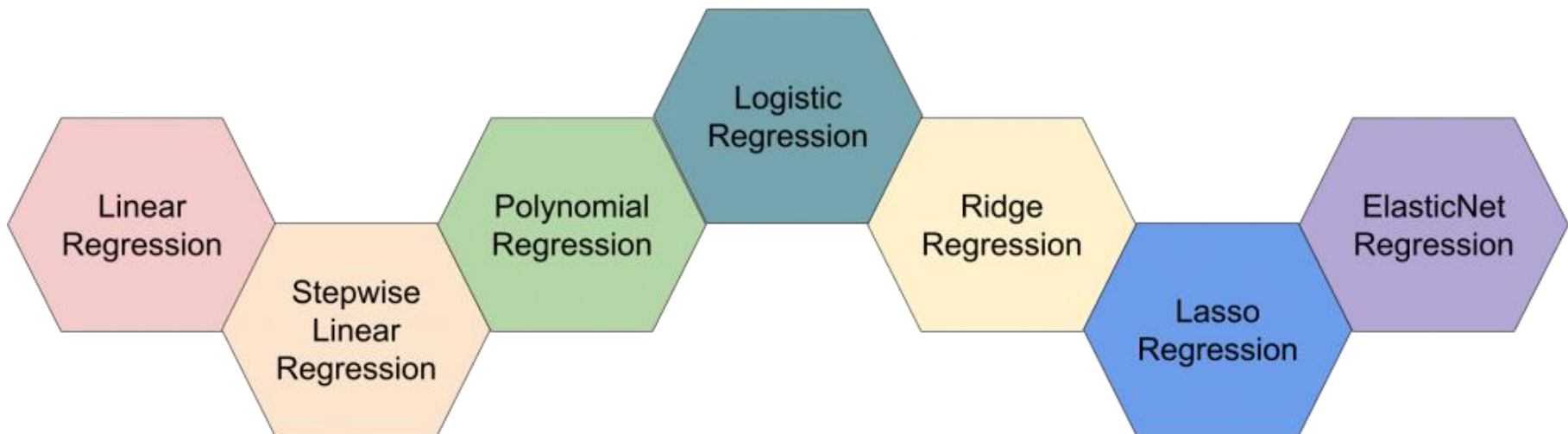
Data Pre-processing/Data Cleaning

- Processing data before it's used to train a model.
- Preprocessing could be as simple as removing **Null values**, **NaN**, **Infinity**, etc
- Filling the missing values.
- Identify outliers and smoothout the data

Machine Learning Basic Terminologies

Regression

- A type of model that **outputs continuous values**.
- A regression problem is when the output variable is a real value, such as “dollars” or “weight”.
- **Regression Algorithms**

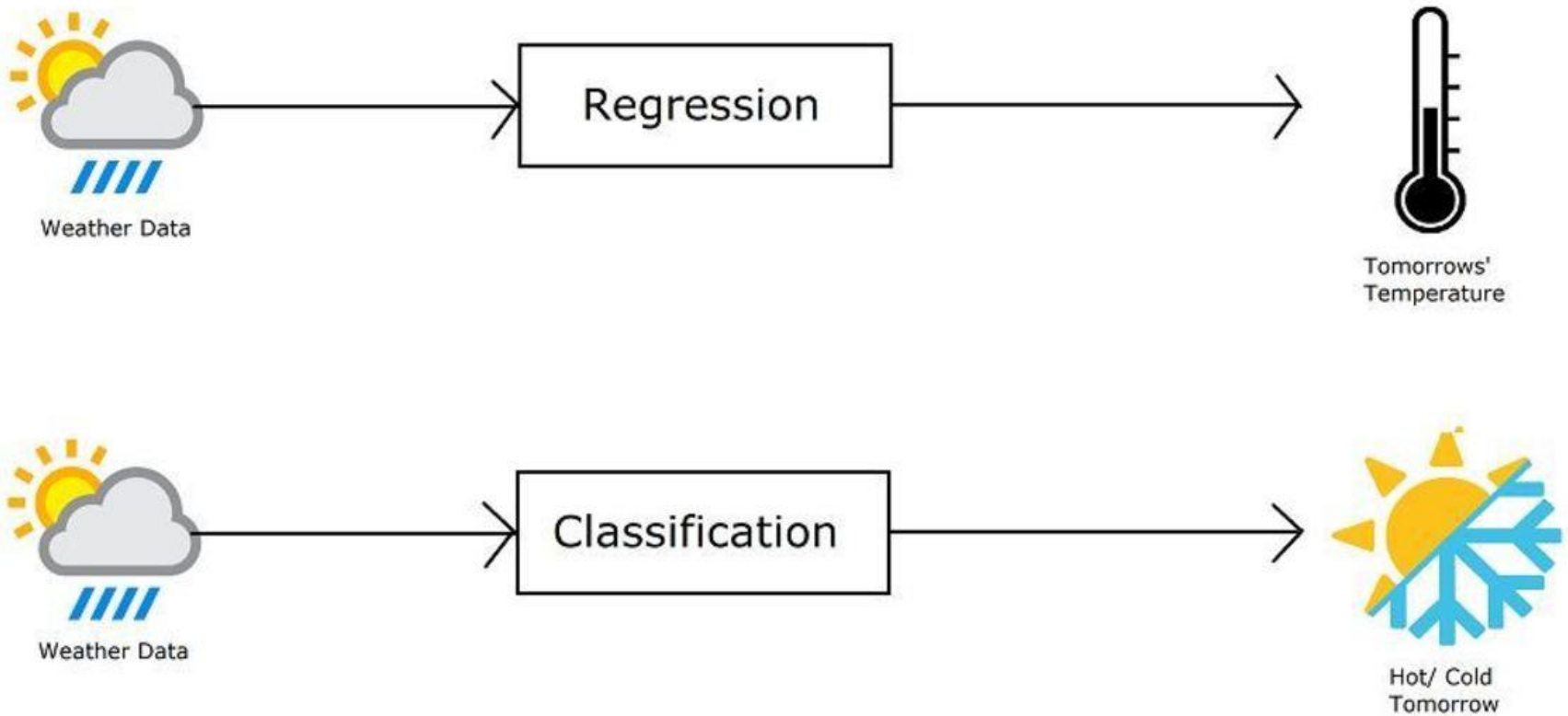


Machine Learning Basic Terminologies

Classification

- A classification problem is when the **output variable is a category**.
- For example, in a **binary classification** model that detects spam, the two classes are spam and not spam. In a **multi-class classification** model that identifies dog breeds, the classes would be poodle, beagle, pug, and so on.
- **Classification Algorithms:**
K-Nearest Neighbors, Support Vector Machine, Decision Tree, Random Forest, Adaboost, etc.

Machine Learning Basic Terminologies



Machine Learning Basic Terminologies

$$Y = w_1x_1 + w_2x_2 + \dots w_nx_n + b$$

Weights

- A coefficient for a feature in a linear model, or an edge in a deep network.
- The goal of training a linear model is to determine the ideal weight for each feature.

Bias

- An intercept or offset from an origin.
- To give importance to some of the features, pre-existing cultural

Machine Learning Basic Terminologies

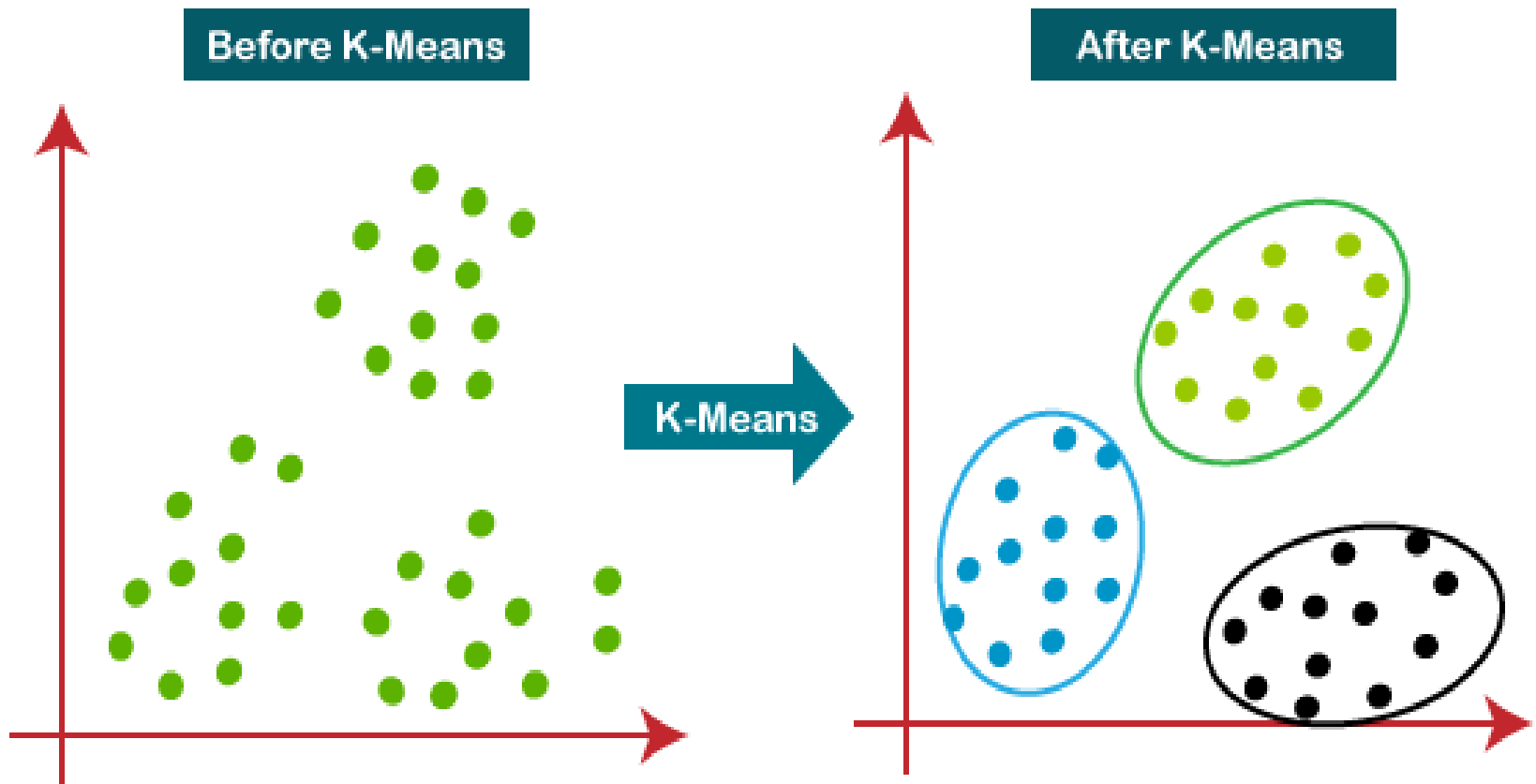
Clustering

- Grouping related examples, particularly during unsupervised learning.
- Once all the examples are grouped, a human can optionally supply meaning to each cluster.
- **Clustering Algorithms:** k-means, Agglomerative Clustering

Centroid

- The center of a cluster as determined by a k-means.
- For instance, if k is 3, then the k-means finds 3 centroids.

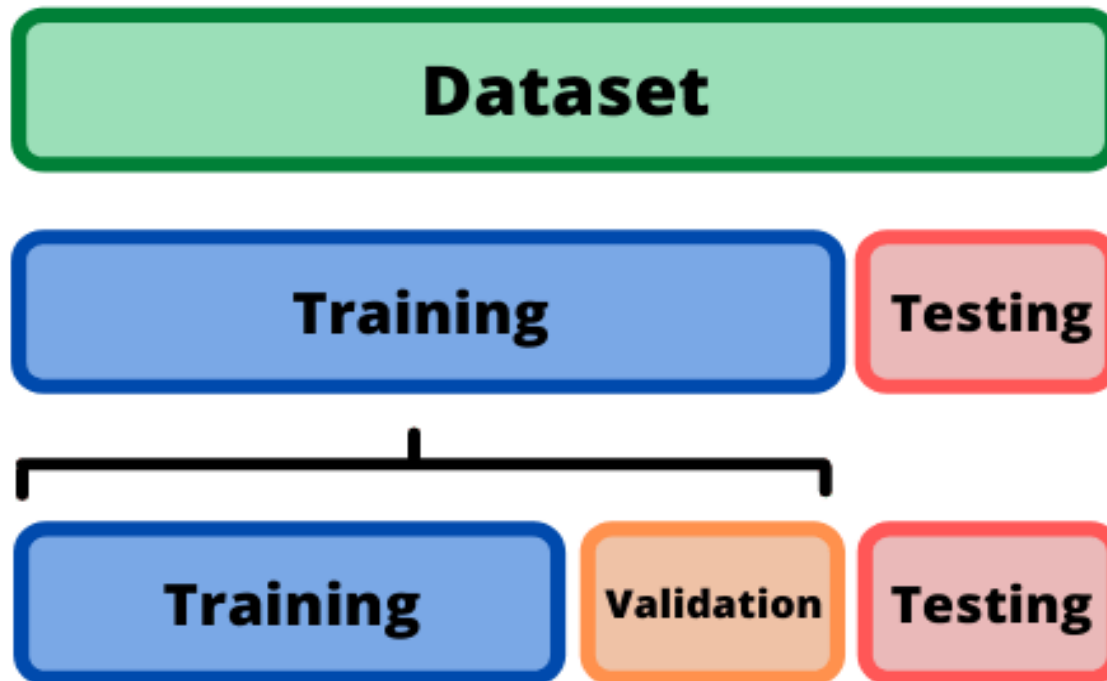
Machine Learning Basic Terminologies



Machine Learning Basic Terminologies

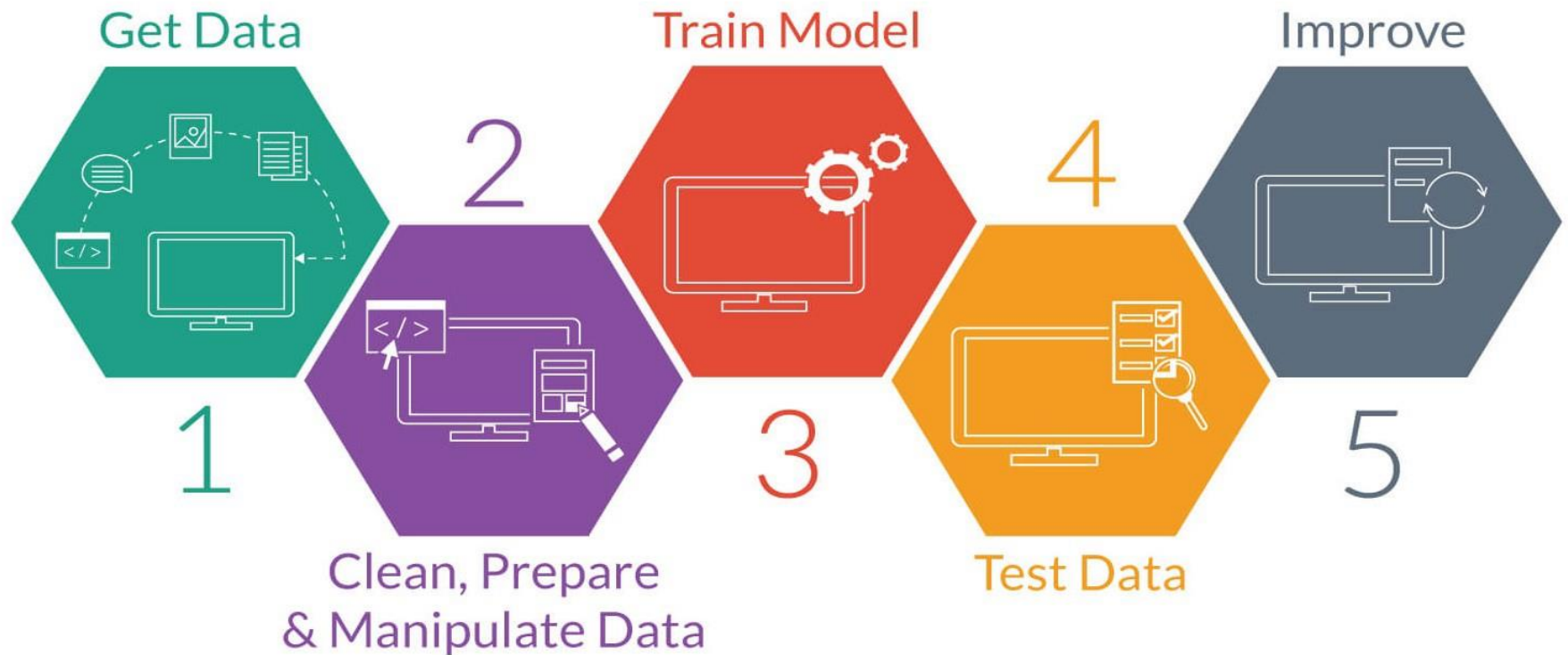
Data Split (Train-set, Test-set, Validation set)

- The subset of the dataset used to train ML model > Train-set



Machine Learning Basic Terminologies

Machine Learning Pipeline



Machine Learning Basic Terminologies

Machine Learning Pipeline

